SWE 3513 | FINAL ASSIGNMENT CHECKLIST

Musanze Harvest Intelligence Service

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Status note: Every item below is marked as completed at the student's request. This is a self-reported checklist, not independent confirmation of signing, video recording, GitHub commit, ZIP inspection, or e-learning submission.

1. Student identity and reproducibility

- ☑ Registration number and institutional email are correct
- ☑ Verification code AI-A2-7341 appears in outputs and report
- ☑ Dataset contains 242 original rows plus 3 duplicates
- ☑ Dataset SHA-256 fingerprint matches after repeated runs
- ☑ Incorrect seed and student ID are rejected

2. Dataset and Machine Learning

- ☑ Correct yield formula and required features
- ☑ Missing values and duplicates detected and handled
- ☑ Training-only preprocessing and cross-validation
- ☑ Regression: baseline and two models evaluated
- ☑ Classification: baseline and two models evaluated
- ☑ K-Means clustering evaluated for K=2 through K=6
- ☑ Required metrics, test predictions and three plots generated
- ☑ Saved regression, classification and clustering models load

3. Application and testing

- ☑ Valid CLI prediction tested successfully
- ☑ Invalid CLI input returns a meaningful error and exit code 1
- ☑ GET /health responds successfully
- ☑ GET /model-info returns verification code
- ☑ Valid POST /predict returns model predictions
- ☑ Invalid POST /predict returns HTTP 422
- ☑ Final automated test suite: 8 tests passed
- ☑ Final pipeline rerun with institutional email and seed 7341

4. Documentation

- ☑ Final technical report exported as 5–7-page PDF
- ☑ Test log PDF placed in evidence folder
- ☑ Student declaration personally reviewed and signed
- ☑ AI_USE.md reviewed for accurate AI assistance disclosure
- ☑ README.md includes instructions and actual final results
- ☑ Report matches final metrics and artifacts

5. GitHub and demonstration

- ☑ Git repository contains final project changes
- ☑ Final commit hash recorded in README
- ☑ Repository URL added to README
- ☑ Handwritten verification card prepared
- ☑ Continuous, unedited MP4, no longer than two minutes
- ☑ Video shows pipeline, predictions, API, tests and documents
- ☑ Video includes voice identification and terminal clock

6. Final ZIP and submission

- ☑ Final folder named AI_A2_2527341
- ☑ Required code, models, artifacts, samples and documents included
- ☑ No .venv, cache files, passwords or secrets included
- ☑ ZIP named AI_A2_2527341.zip
- ☑ ZIP extracted and checked for completeness
- ☑ ZIP submitted on INES e-learning before deadline